Checking the origin of the Asymmetry of the Surface Detector signals

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1 Introduction

A forward-backward asymmetry is expected on the integral and the shape of the signals in the Cherenkov tanks of the Surface Detector: it is predicted by the simulation of the detector response to inclined showers, and it is observed on the real events observed with the Engineering Array. The effect is a combination of the longitudinal evolution of the shower and a geometrical effect related to the incidence angles of the particles on the walls of the tanks; it was studied in detail in [1], where the geometrical effect is shown to be dominant at moderate zenith angles, giving a large asymmetry over short distances, much larger than the intrinsic longitudinal attenuation of the shower.

In this study we want to illustrate the phenomenon through a simulation of tanks of different shapes, to modify the geometrical factors without changing the shower itself: the first shape is the "normal" one (diameter much larger than height), while the second is higher than wide; in other terms, the ratio of lateral to top area is reversed. This influences the response to *electromagnetic* particles (e^-, e^+, γ), but practically not the response to *muons*, which produce light throughout the volume of the water.

2 The geometrical effect

The particles in the shower, in average, diverge from the axis: then the flux received in the tanks on the "upstream" side (or "early" tanks) is more vertical flux than in the "downstream" ones ("late" tanks). Quantitatively, if we call α the divergence angle, $\mathcal{A}_{top}$ and $\mathcal{A}_{side}$ the top and the lateral areas, we can define an effective area which is $\mathcal{A}_{top}\cos(\theta-\alpha)+\mathcal{A}_{side}\sin(\theta-\alpha)$ for "early" tanks, and $\mathcal{A}_{top}\cos(\theta+\alpha)+\mathcal{A}_{side}\sin(\theta+\alpha)$ for "late" tanks (we assume these tanks to lie on the longitudinal axis to avoid mathematical complications). Then, for a given intrinsic density in the shower, flat-shaped tanks (with a top area much larger than the side wall area) receive more particles in the upstream region, while high-shaped tanks receive more in the downstream region (see Fig.1). This induces the *geometrical* asymmetry, with a sign depending on the shape of the tanks. With the present radius/height ratio of 1.5, its magnitude may be of ± 30 %, even at a distance r from axis as low as 500 m.

The Cherenkov light deposited by muons is roughly proportional to their path in water, and they have generally enough energy to go through the tank; as a consequence,

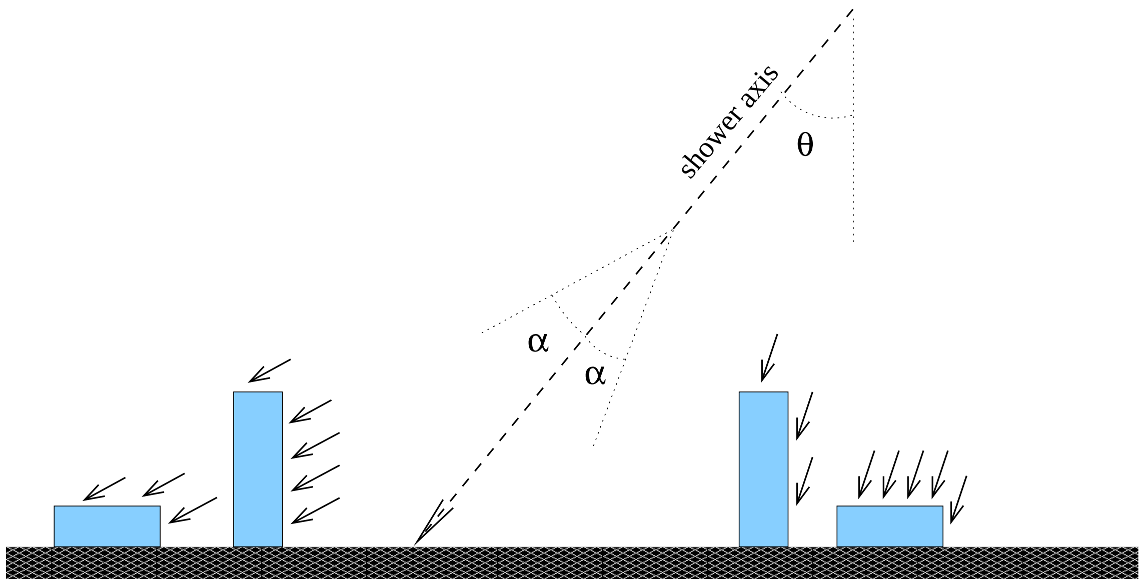

Figure 1: The geometrical forward-backward asymmetry for photons and electrons.

the response to a given flux is proportional to the *volume* of the tank, whatever the incident direction: there is no geometrical asymmetry from muons.

Of course, this effect is combined with the longitudinal evolution of the shower over a length $2r\sin\theta$ (between an "early" and a "late" tank, both at distance r from the axis). At low zenith angles (up to $\theta \sim 30$ deg), at a distance around 1000 m, this evolution is not important; depending on the primary particle (and the interaction model) it may be positive or negative. At larger θ, it consists in an attenuation for both electromagnetic and muonic components, with a characteristic length of 5 to 10 km, that is less than 10 % with $r = 500$ m and $\theta = 30$ deg.

Accounting for all components, and knowing that the muonic/electromagnetic ratio increases with θ and r, we expect:

- for flat-shaped tanks (actual shape): more signal in "early" tanks at any value of θ, with an asymmetry going to 0 when $\theta \to 0$ and to low values for horizontal showers (dominated by muons).
- for high-shaped tanks: a reversed asymmetry at moderate values of θ, where the geometrical effect is dominant, but the same asymptotical behaviour as flat-shaped tanks at large θ.

3 Results of the simulation

We have used a subsample of the AIRES shower library produced at CCIN2P3 at Lyon (version 2.5, with QGSJET01) at energies ranging from 1 to 100 EeV, and $\cos\theta$ ranging from 0.1 to 1 by steps of 0.1. The ground particle files were input to the detector simulation package *sample_sim*, with a set of tanks defined *ad hoc* (distance to core from 200 to 2500 m, 18 values regularly spaced in azimuthal angle ψ in frontal projection). All simulations were performed twice, once with the nominal shape (radius 1.8 m, water height 1.2 m) and once with another shape for the same total volume (radius 0.9 m, height 4.8 m). The signal, as a function of ψ, is given on Fig. 2: it exhibits the expected behaviour, especially the reversed asymmetry at low zenith angle, and a "saturation" at large distance (a pure attenuation effect would be roughly proportional to the distance).

Note that the dependence in ψ is not exactly in $(1 + k\cos\psi)$, because the geometric factor is more complex in 3D than in our simplified approach, and the longitudinal attenuation is not linear; however, when the asymmetry is large, this approximation is adequate for the actual shape of the tanks, with respect to the intrinsic fluctuations of the signal.

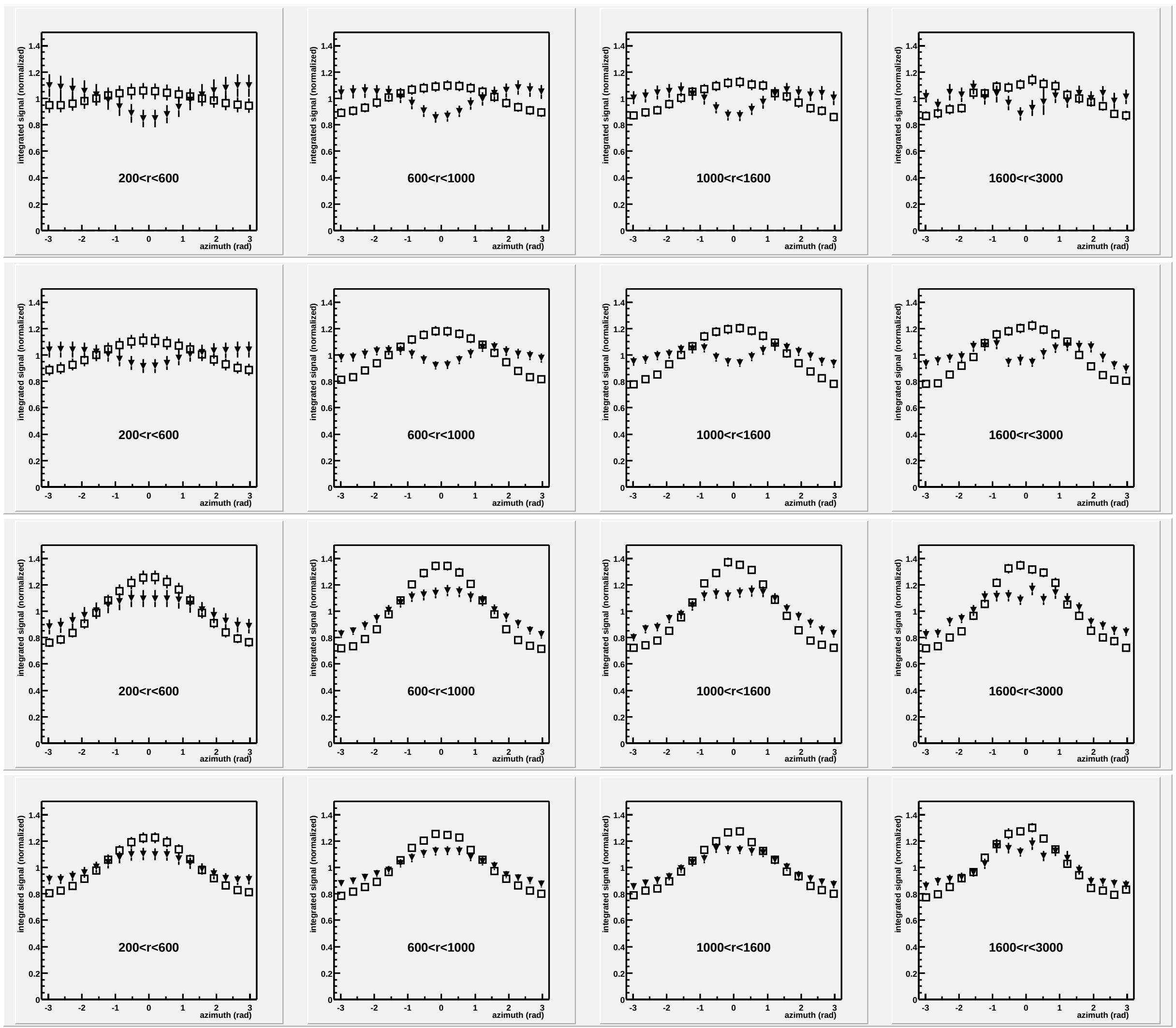

Figure 2: The azimuthal dependence of the integrated response at $\theta = 25, 36, 53, 66$ deg (from top to bottom), at different distances from shower axis, for normal-shaped tanks (white squares) and high-shaped tanks (black triangles); the averaged signal is normalized to the same value.

4 Conclusion

The simulation of water tanks with different shapes gives an evidence for the different contributions to the forward-backward asymmetry of the detector response, especially the domination at low zenith angle of the geometric effect on the electromagnetic component; this effect, due to the divergence of the particles from the shower core. is related to the shape of the tank, and not to the longitudinal evolution of the particle densities in the shower.

References

[1] X. Bertou, P. Billoir: On the Origin of the Asymmetry of Ground Densities in Inclined Showers (GAP-2000-017)